SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	S.PRATEEKSHA	**Reg. No.:**	192572137
Date:	31/7/26	**Duration:**	60 minutes

Code of Conduct

I S.PRATEEKSHA-192572137 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S.PRATEEKSHA

Business domain:

Employee performance evaluations and productivity analytics system

Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.

Project Vision

The Employee Performance Evaluations and Productivity Analytics System is designed to provide an efficient and transparent platform for managing employee performance evaluations and analyzing productivity-related information. The system will allow managers and HR personnel to set performance criteria, conduct periodic evaluations, monitor objectives, view productivity trends, and generate reports. Employees will be able to view their goals, performance feedback, and evaluation results.

The main vision of the project is to replace manual and time-consuming performance evaluation processes with a centralized, data-driven, and user-friendly system that supports fair performance assessment and better organizational decision-making.

Objectives

1. Automate employee performance evaluation and feedback processes.
2. Maintain employee goals, targets, and performance records in a centralized system.
3. Provide productivity analytics through dashboards and reports.
4. Help managers identify strengths, weaknesses, and areas for improvement.
5. Provide HR and management with useful data for performance-related decisions.

6. Improve transparency by allowing employees to view their evaluation results and feedback.

Stakeholder	Role / Responsibility
Employees	View goals, submit self-evaluations, and view feedback and performance results.
Managers / Team Leaders	Set goals, evaluate employees, provide feedback, and monitor team productivity.
HR Managers	Define evaluation policies, monitor employee performance, manage evaluation cycles, and generate reports.
Senior Management	View organizational-level performance and productivity analytics for decision-making.
System Administrator	Manage users, roles, permissions, and system configuration.
Project Team / Developers	Analyze, design, develop, test, and maintain the system.
Product Owner	Represents business needs, prioritizes the product backlog, and validates completed features.
QA / Testers	Verify that the system satisfies functional and quality requirements.

2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format:
     As a <user>, I want <feature>, so that <benefit>.

Epic 1 – User Authentication and Role Management

US1:

As an **employee**, I want to log in securely, so that I can access my performance information.

US2:

As a **system administrator**, I want to assign user roles, so that employees, managers, and HR users receive appropriate access.

Epic 2 – Employee Profile and Goal Management

US3:

As a **manager**, I want to create performance goals for employees, so that their expected targets are clearly defined.

US4:

As an **employee**, I want to view my assigned goals, so that I understand what I am expected to achieve.

Epic 3 – Performance Evaluation

US5:

As a **manager**, I want to evaluate employees using predefined performance criteria, so that employee performance can be measured consistently.

US6:

As an **employee**, I want to submit a self-evaluation, so that I can provide my perspective on my performance.

US7:

As an **employee**, I want to view my evaluation results and feedback, so that I can understand my strengths and areas for improvement.

Epic 4 – Productivity Analytics

US8:

As a **manager**, I want to view productivity metrics for my team, so that I can identify performance trends and areas requiring attention.

US9:

As an **HR manager**, I want to compare performance trends across teams, so that I can identify organizational-level patterns.

Epic 5 – Reports and Dashboard

US10:

As an **HR manager**, I want to generate performance reports, so that I can support performance reviews and management decisions.

US11:

As a **senior manager**, I want to view a performance dashboard, so that I can quickly understand overall employee productivity and performance trends.

Epic 6 – Notifications and Feedback

US12:

As an **employee**, I want to receive notifications when an evaluation is completed, so that I know when new feedback is available

3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.

The following acceptance criteria are defined using the **Given–When–Then** format for the high-priority user stories.

US1 – Secure Login

Given the user has a valid registered account
When the user enters the correct username and password
Then the system should authenticate the user and display the appropriate dashboard.

Given the user enters incorrect credentials
When the login button is selected
Then the system should display an appropriate error message and deny access

US2 – Role Management

Given the administrator is logged in
When the administrator assigns a role to a user
Then the system should save the selected role and apply the corresponding permissions.

Given a user has an employee role
When the user accesses the system
Then restricted manager or administrator functions should not be accessible

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US3 – Create Employee Goals

Given a manager is logged in

When the manager enters an employee goal, target, and due date

Then the system should save the goal and associate it with the selected employee.

Given required goal information is missing

When the manager attempts to save the goal

Then the system should display a validation message

US4 – View Assigned Goals

Given an employee has assigned goals

When the employee opens the Goals section

Then the system should display all active goals, targets, and deadlines.

US5 – Employee Performance Evaluation

Given an evaluation period is active

When a manager selects an employee and enters scores for all required performance criteria

Then the system should calculate and save the overall evaluation result.

Given mandatory criteria are incomplete

When the manager submits the evaluation

Then the system should prevent submission and identify the missing criteria.

US6 – Self-Evaluation

Given a self-evaluation period is open

When an employee enters ratings and comments

Then the system should save and submit the self-evaluation.

US7 – View Feedback

Given a manager has completed an employee evaluation
When the employee opens the performance results page
Then the employee should be able to view the evaluation score and manager feedback.

US8 – Productivity Analytics

Given productivity data is available
When a manager opens the analytics dashboard
Then the system should display relevant productivity metrics and performance trends.

Given there is insufficient data for a selected period
When the manager requests analytics
Then the system should clearly indicate that sufficient data is unavailable

4. **Prioritize the Product Backlog**

   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.The Product Backlog is prioritized using **business value, user importance, dependencies, and implementation needs**.

The product backlog for the **Employee Performance Evaluations and Productivity Analytics System** is prioritized based on business value, user requirements, technical dependencies, and the importance of each feature to the overall system. The features that are essential for building the basic system are given the highest priority, while advanced features are planned for later sprints.

Priority 1 – Highest Priority

1. User Login and Authentication (US1)
This feature is given the highest priority because all users must securely log in before accessing the system. It provides the basic security foundation of the application.

2. Role Management (US2)

Role management is prioritized next because different users such as employees, managers, HR managers, and administrators require different levels of access.

3. Create Employee Goals (US3)

Managers need to create and assign goals to employees before their performance can be evaluated. Therefore, this is an important core functionality.

4. View Assigned Goals (US4)

Employees must be able to view their assigned goals, targets, and deadlines. This feature depends on the goal creation functionality.

Priority 2 – High Priority

5. Employee Performance Evaluation (US5)

This is the main functionality of the system. Managers can evaluate employees based on predefined performance criteria and provide performance scores.

6. Employee Self-Evaluation (US6)

This allows employees to assess their own performance and provide comments, making the evaluation process more participatory and transparent.

7. View Evaluation Feedback (US7)

Employees should be able to view their performance scores and manager feedback so that they can understand their performance and areas for improvement.

8. Productivity Analytics (US8)

This feature provides managers with productivity metrics and performance trends. It is prioritized after the basic performance data is available.

9. Performance Report Generation (US10)

HR managers can generate performance reports for employees or teams. These reports support management decisions and performance reviews.

Priority 3 – Medium Priority

10. Management Performance Dashboard (US11)

A dashboard provides senior management with a summarized view of employee performance and productivity trends. It is useful but depends on sufficient performance data.

11. Notification System (US12)

Notifications inform employees and managers about evaluation deadlines, completed evaluations, and feedback availability. This improves communication but is not essential for the initial system.

12. Team Comparison Analytics (US9)

This advanced feature allows HR managers to compare performance across different teams and identify organizational trends. It can be implemented after the core analytics features are completed.

Prioritization Summary

- The development should begin with **authentication and role management**, followed by **goal management and performance evaluation**. Once sufficient performance data is collected, **analytics, reports, dashboards, and notifications** can be developed. This priority order reduces dependencies and ensures that the most valuable features are delivered first.

5.Plan the Sprint Backlog

- Select high-priority user stories for the first sprint and divide them into actionable development tasks.

For the first sprint, the highest-priority user stories are selected based on their importance, dependencies, and feasibility within the sprint duration. The first sprint focuses on building the basic foundation of the **Employee Performance Evaluations and Productivity Analytics System**.

Sprint 1 Selected User Stories

US1 – User Login and Authentication

As an employee, I want to log in securely, so that I can access my performance information.

Development Tasks:

- Design the login page.
- Create the user authentication database.
- Implement username and password validation.
- Handle invalid login attempts.
- Test successful and unsuccessful login scenarios.

US2 – Role Management

As a system administrator, I want to assign user roles, so that users can access only the functions allowed for their roles.

Development Tasks:

- Define employee, manager, HR, and administrator roles.
- Create role and permission structures.
- Implement role assignment.
- Restrict access based on user roles.
- Test role-based access.

US3 – Create Employee Goals

As a manager, I want to create performance goals for employees, so that their expected targets are clearly defined.

Development Tasks:

- Design the goal creation interface.
- Create the goal database.
- Add employee goal details.
- Implement target and deadline validation.
- Save goals in the database.

- Test goal creation.

US4 – View Assigned Goals

As an employee, I want to view my assigned goals, so that I understand what I am expected to achieve.

Development Tasks:

- Design the employee goal page.
- Retrieve assigned goals from the database.
- Display targets and deadlines.
- Display goal status.
- Test goal viewing functionality.

At the end of Sprint 1, these features should work together as a basic functional version of the system

5. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.

The **Fibonacci sequence** is used for estimating the effort required for each user story. The values used are **1, 2, 3, 5, 8, 13**, where a higher number represents greater complexity, effort, uncertainty, or dependency.

US1 – User Login and Authentication: 3 Story Points

This story requires the development of a login interface, credential validation, database interaction, and basic security handling. The functionality is relatively straightforward, so it is estimated at **3 points**.

US2 – Role Management: 3 Story Points

Role management involves creating different user roles and controlling access to system features. Since the number of roles is limited and the functionality is manageable, it is assigned **3 points**.

US3 – Create Employee Goals: 5 Story Points

This story requires a user interface, database operations, validation, and association between employees and their goals. Because it contains multiple components, it is estimated at **5 points**.

US4 – View Assigned Goals: 3 Story Points

This functionality mainly involves retrieving stored goals and displaying them to employees. The implementation is less complex compared with goal creation, so it is assigned **3 points**.

Total Sprint Effort

- The total estimated effort for Sprint 1 is:

  3 + 3 + 5 + 3 = 14 Story Points

- The estimates can be finalized using **Planning Poker**, where team members independently provide estimates and discuss differences before reaching a common estimate.
- The **Fibonacci sequence (1, 2, 3, 5, 8, 13...)** is used for estimation. Story points represent the relative combination of **complexity, effort, uncertainty, and dependencies**, rather than exact hours.

7.Present the Sprint Goal and Expected Deliverables

- Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

print Goal

- To develop the basic foundation of the Employee Performance Evaluations and Productivity Analytics System by implementing secure login, role-based access, employee goal creation, and goal viewing functionalities.
- The sprint focuses on creating a working foundation that can support the development of performance evaluation and productivity analytics features in future sprints.

Expected Deliverables

1. Working Login Module

Users can securely log in using valid credentials and unauthorized users are prevented from accessing the system.

2. Role-Based Access Module

The system provides different access permissions for employees, managers, HR managers, and administrators.

3. Employee Goal Management Module

Managers can create and assign goals to employees with appropriate targets and deadlines.

4. Employee Goal Viewing Module

Employees can view their assigned goals, targets, deadlines, and status.

5. Basic Database Structure

The initial database required for users, roles, employees, and goals will be created and connected to the application.

6. Tested Software Increment

All completed features will be integrated and tested to ensure that the Sprint 1 version works correctly.

Sprint Completion

At the end of the sprint, the team should have a **working prototype of the core system**, which can be demonstrated to the Product Owner and used as the foundation for the next sprint. The subsequent sprint can focus on **performance evaluation, self-evaluation, feedback, and productivity analytics**.

Conclusion

- The Agile Sprint Planning and User Story Workshop for the **Employee Performance Evaluations and Productivity Analytics System** provides a structured approach for developing the project step by step.
- The system requirements were divided into epics and user stories, with clear acceptance criteria and priorities based on business value and dependencies. For Sprint 1, the focus is on developing the basic foundation through secure login, role management, employee goal creation, and goal viewing. Using Fibonacci-based story point estimation helps the development team understand the effort and complexity of each feature.
- Overall, the Agile approach supports continuous development, testing, feedback, and improvement while ensuring that important employee performance and productivity features are delivered effectively.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint	Story point estimation is well justified, sprint goal is clear, deliverables are	Estimation and sprint goal are mostly appropriate with minor	Estimation or sprint goal lacks justification.	Estimation is unrealistic or missing. Sprint

Goal & Presentation	realistic, and presentation is professional and well organized.	inconsistencies. Presentation is clear.	Presentation is adequate but incomplete.	goal, deliverables, and presentation are unclear or incomplete.